CSE221: Algorithms

	Course Contents	Chapter Notes	My Recordings	Other Resources	Practice Questions
MID SYLLABUS	Introduction to Algorithms	CSE221 Chap 0.pdf	Intro to Algorithms		
	Asymptotic Analysis - Big Oh, O - Big Omega, Ω - Big Theta, θ	CSE221 Chap 1.pdf	Asymptotic Analysis (Part 1) Asymptotic Analysis (Part 2)		Asymptotic Analysis & Time Complexity
	Time Complexity - Simplified Rules - Series Rules - Iterative Time Complexities - Recursive Time Complexities	CSE221 Chap 2.pdf	Time Complexity (Part 1)	Advanced Master Theorem Time Complexity Part 1 Time Complexity Part 2 Time Complexity Part 3 Time Complexity Part 4	Asymptotic Analysis & Time Complexity
	Searching Algorithms - Linear Search - Binary Search - Ternary Search	CSE221 Chap 3.pdf	Ternary Search		Searching Algorithms
	Sorting Algorithms - Merge Sort - Quick Sort - Count Sort	CSE221 Chap 4.pdf	Merge and Quick Sort		Sorting Algorithms
	Divide-and-Conquer - Maximum Subarray Problem - Karatsuba's Algorithm	CSE221 Chap 5.pdf	Karatsuba's Multiplication	Maximum Subarray Karatsuba's Multiplication	Divide-and-Conquer
	Graph Traversal 1 - BFS - BFS Applications	CSE221 Chap 6.pdf	BFS and BFS Applications		Graph Traversal (BFS)
	Graph Traversal 2 - DFS - DFS Applications	CSE221 Chap 6.pdf	DFS and DFS Applications		Graph Traversal (DFS) & Graph Algorithms
FINAL SYLLABUS	Graph Algorithms - Topological Sort - Strongly Connected Components - Shortest Path (Dijkstra)	CSE221 Chap 7.pdf		Dijkstra's Algorithm	Graph Traversal (DFS) & Graph Algorithms
	Greedy Algorithms - Spanning & Minimum Spanning Tree - Kruskal's Algorithm - Huffman Coding - Interval Scheduling	CSE221 Chap 8.pdf CSE221 Chap 8.1.pdf		Huffman Coding	Greedy Algorithms
	Dynamic Programming - DP Basics - Knapsack Problem (Fractional & 0/1) - Longest Common Subsequence (LCS)	CSE221 Chap 9.pdf		Intro to DP 0/1 Knapsack LCS	Dynamic Programming
	Backtracking Algorithms - 15 Puzzle Problem	CSE221 Chap 10.pdf			